

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7649
- **Team Leader:** N.Sopya

2. LEVEL 0 DATA FLOW DIAGRAM (CONTEXT DIAGRAM) The Context Diagram provides a high-level view of the entire ComicCraft system.

- **External Entity (User):** Submits [Story Prompt, Character Name, Tone, Style, Setting] to the ComicCraft System.
- **ComicCraft System (Main Process):** Processes the inputs, coordinates with external AI services, and returns a [Rendered HTML Preview] and a [Downloadable PDF] back to the User.
- **External Entity (Google Gemini AI):** Receives the [Text Prompt] from the system and returns a structured [Panel Outline & Narration].
- **External Entity (Hugging Face API):** Receives the [Image Prompt] from the system and returns [Generated Image Data].

3. LEVEL 1 DATA FLOW DIAGRAM (SYSTEM BREAKDOWN) This level breaks down the main process into specific functional sub-processes.

- **Process 1.0 (Input Collection):**
 - Captures user data from the web form via the /generate POST route.
 - *Data Out:* Validated Form Data mapping.
- **Process 2.0 (Story Structuring):**
 - Sends data to Gemini Flash.
 - *Data In:* Validated User Input.
 - *Data Out:* Structured 5-Panel Outline (JSON).
- **Process 3.0 (Dialogue Expansion):**
 - Sends the generated outline to Gemini Pro.
 - *Data In:* 5-Panel Outline.

- *Data Out*: Full Comic Narration Text and Dialogue.
- **Process 4.0 (Image Generation):**
 - Iterates through the outline and queries the Stable Diffusion model.
 - *Data In*: AI-generated Image Prompts.
 - *Data Out*: Saved .jpg images in the local `static/panels/` directory.
- **Process 5.0 (Layout & Compilation):**
 - Merges the generated Text and Images into a unified data structure.
 - *Data In*: Narration Text, Scene Descriptions, Image Paths.
 - *Data Out*: Comprehensive Layout Dictionary.
- **Process 6.0 (PDF Export):**
 - Converts the Layout Dictionary into a multi-page PDF using the FPDF module.
 - *Data Out*: Downloadable `comic_timestamp.pdf`.